Are post-Middle Miocene tilts present in the southern Coast Belt of British Columbia? A case study of present Earth's magnetic field overprints[1]

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Tectonic tilt is a possible explanation for the discordant Cretaceous paleomagnetic poles obtained from intrusions in the Coast Belt of the Canadian Cordillera. If tilting of these intrusions has taken place, post-Cretaceous intrusions may have been similarly tilted. Paleomagnetism of the Middle Miocene Rogers Creek pluton (16 Ma) in the central-southern Coast Belt was utilized to test the tilt hypothesis. Well-grouped stable primary magnetizations have been isolated in 11 sites with both normal and reverse polarities. The mean remanence component, directed at $D = 000.7°$, $I = 67.9°$ ($\alpha_{95} = 3.9°$, paleolatitude 50.9 ± 9°N, paleopole 88.3°N, 257.5°E, $A_{95} = 5.7°$), was likely acquired upon emplacement at 16 Ma. This direction agrees with the Middle Miocene reference direction ($D = 357°$, $I = 68°$) calculated from the updated Miocene (10–20 Ma) cratonic reference pole for North America. This concordant result, when combined with concordant data from other mid-Tertiary intrusions to the south, defines a southern Coast Belt domain that shows no evidence of post-Middle Miocene tectonic tilt. This result does not preclude tilting of the mid-Cretaceous Spuzzum batholith within this domain; however, if tilting took place it must have occurred prior to the Middle Miocene. The concordant data from the Rogers Creek pluton provide a Middle Miocene paleohorizontal datum for future work investigating crustal tilts in the Coast Belt.

A comparison of Tertiary Coast Belt paleopoles demonstrates that they are systematically rotated clockwise and are near-sided relative to the cratonic reference poles. This implies that the Coast Belt has been displaced 300 km southward and rotated 8° clockwise, or regionally tilted downward to the northwest, or some combination of these two end members, since the Late Miocene. These scenarios appear untenable. Evidence is given that the directional bias of the paleopoles is caused by unremoved present Earth's magnetic field (PEMF) overprint magnetizations, which comprise 10–25% of the total remanence in most cases. The implication is that demagnetization and analysis techniques used in previous Tertiary Coast Belt studies, to varying degrees, may have insufficiently removed the PEMF overprints. The case is made for the importance of thorough alternating field and thermal step demagnetization, of all specimens, in order to maximize magnetic component data from which the principal magnetic components can be best isolated and PEMF components removed.

Un basculement tectonique peut être envisagé pour expliquer les pôles paléomagnétiques discordants crétacés, fournis par les intrusifs du Domaine côtier de la Cordillère canadienne. Si le basculement tectonique de ces intrusifs s'est produit, il est probable que les intrusifs postcrétacés aient pu aussi être dérangés de façon analogue. Une étude paléomagnétique des roches du pluton de Rogers Creek, d'âge miocène moyen (16 Ma), dans le centre-sud du Domaine côtier, a été effectuée pour tester l'hypothèse d'un tel basculement. Des aimantations primaires stables, bien regroupées, ont été isolées dans 11 sites, elles présentent des polarités normales et inverses. La composante de rémanence moyenne, de direction $D = 000{,}7°$, $I = 67{,}9°$ ($\alpha_{95} = 3{,}9°$, paléolatitude 50,9 ± 9°N, paléopôle 88,3°N, 257,5°E, $A_{95} = 5{,}7°$), fut acquise vraisemblablement lors de la mise en place, il y a 16 Ma. Cette direction est en accord avec la direction référentielle ($D = 357°$, $I = 68°$) pour le Miocène moyen, calculée à partir du pôle de référence cratonique de l'Amérique du Nord qui a été mis à jour pour le Miocène (10–20 Ma). Ce résultat concordant, lorsque combiné aux données concordantes obtenues pour d'autres intrusifs du milieu du Tertiaire au sud, définit une zone méridionale dans le Domaine côtier qui ne montre aucun des indices d'un basculement tectonique postmiocène moyen. Ce résultat n'élimine pas la possibilité d'un basculement du batholite de Spuzzum, d'âge crétacé moyen, au sein de cette zone méridionale; cependant, s'il y a eu basculement, il a dû se produire antérieurement au Miocène moyen. Les données concordantes obtenues pour le pluton de Rogers Creek fournissent un paléohorizontale datée du Miocène moyen à être utilisée pour les études futures des basculements crustaux dans le Domaine côtier.

Une comparaison des paléopôles tertiaires du Domaine côtier démontre qu'ils sont systématiquement tournés dans le sens horaire et rapprochés relativement aux pôles cratoniques de référence. Ce qui implique un déplacement du Domaine côtier de 300 km vers le sud et une rotation de 8° dans le sens des aiguilles d'une montre, ou un basculement régional avec abaissement vers le nord-ouest, ou une certaine combinaison de ces deux types de mouvement, depuis le Miocène tardif. Il est difficile de plaider en faveur de ces scénarios. Des indices suggèrent que la divergence de la direction des paléopôles résulte du fait que la surimpression des aimantations du champ magnétique terrestre actuel n'a pas été corrigée, ce qui représente dans certains cas 10–25% de la rémanence totale. Par conséquent, les techniques de désaimantation et d'analyses utilisées dans les études antécédentes du Domaine côtier tertiaire, jusqu'à un certain degré, ne supprimaient pas suffisamment les surimpressions du champ magnétique actuel. L'étude du présent cas démontre l'importance d'une désaimantion complète par champ alternatif et par chauffage, de tous les spécimens, afin d'optimiser les données des composantes magnétiques, à partir desquelles les principales composantes magnétiques seraient le mieux isolées, et afin de supprimer suffisamment les composantes du champ magnétique terrestre actuel.

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Introduction

In the Canadian Cordillera paleomagnetically determined Eocene or post-Eocene tilts have been described from the Omineca Belt in southeast British Columbia (Marquis and Irving 1990), from the Insular Belt in the Queen Charlotte Islands (Irving *et al.* 1992), from the Coast Belt (Symons 1977*b*), and from Vancouver Island (Irving and Massey 1990). It is likely that these inferred tilts are tectonic in origin. Alternatively the discordant magnetizations observed in these bodies can be explained by a combination of very large rotations and translation, which seems unlikely, as it is well documented that the terranes involved were in place relative to North America by Eocene time (Bardoux and Irving 1989; Symons and Wellings 1989; Irving and Brandon 1990; Vandall and Palmer 1990*a*). In contrast, numerous other paleomagnetic studies of Tertiary rocks from the Coast Belt (e.g., Beske *et al.* 1973; Symons 1973; Beck *et al.* 1982) are concordant and do not exhibit evidence of tilt. Older Cretaceous magnetizations in rocks of the Coast Belt (Symons 1974, 1977*a*, 1977*b*; Irving *et al.* 1985) are discordant, and post-mid-Cretaceous tilting of these rock units has been inferred (e.g., Symons 1977*a*; Butler *et al.* 1989; Brown and Burmester 1991). However, the alternative or contemporaneous hypotheses involving rotation and translation (e.g., Beck *et al.* 1981; Irving *et al.* 1985; Irving and Wynne 1990; Vandall 1990; Vandall and Palmer 1990*b*) cannot be ruled out because the position of the Coast Belt relative to North America at that time has not yet been firmly established.

The Coast Belt (Fig. 1) of the Canadian Cordillera embodies the core of the longest mountain chain in North America. It still remains an enigma. Tectonic models that seek to explain its origin focus on the style and timing of assembly of the Insular Belt and the Intermontane Belt. For the last decade it has been widely accepted that the Coast Belt formed as a result of mid-Cretaceous collision of the exotic Insular superterrane against the Intermontane superterrane (Davis *et al.* 1978; Monger *et al.* 1982). Prior to this model, Monger *et al.* (1972) described the Coast Belt as a deeply eroded root of a Tertiary magmatic arc that formed as a result of Andean type subduction. As a result of recent stratigraphic, paleontologic, geochronologic, and paleomagnetic data (e.g., Mortimer 1986; van der Heyden 1989; Vandall and Palmer 1990*b*; Rusmore and Woodsworth 1991), van der Heyden (1992) has revived the Andean subduction model to explain the origin of the Coast Belt. In this model van der Heyden (1992) describes the regional tectonic features of the western Canadian Cordillera in terms of Middle Jurassic accretion of a single composite superterrane to ancestral North America. The accretion of this superterrane, comprising Stikinia, Wrangellia, and the Alexander terrane, closed the intervening Cache Creek – Bridge River ocean. This initiated the development of the Coast Belt as a succession of superimposed Middle Jurassic to early Tertiary magmatic arcs related to prolonged subduction of Pacific Ocean lithosphere beneath the new continental margin. This model proposes pre-Middle Jurassic linkages between the Insular and Intermontane superterranes, which have been viewed as widely separate entities that predate mid-Cretaceous time.

How does the discordant Cretaceous Coast Belt paleomagnetic data fit this model? A recent review of the Cretaceous Canadian Cordilleran paleomagnetic data (Vandall 1990) demonstrates that a face-value interpretation of the Cretaceous paleolatitudes determined from these studies would imply that there was as much as 1000 km of differential, post-mid-Cretaceous, northward translation between the Coast and Intermontane

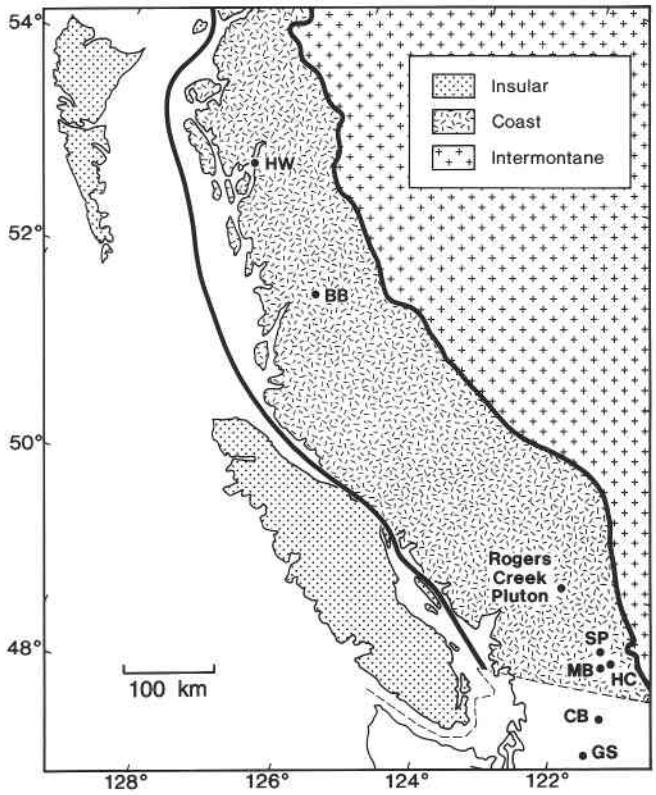


FIG. 1. General tectonic setting of the Rogers Creek pluton within the southern Coast Belt and previous Tertiary Coast Belt paleomagnetic studies including the Cretaceous Spuzzum batholith (SP). BB, Brown basalt dikes; MB, Mount Barr pluton; GS, Grotto and Snoqualmie batholiths; CB, Chilliwack batholith; HC, Hope Complex; and HW, north-northwest Hawksbury warp plutons. Terrane boundaries after Wheeler *et al.* (1991).

belts. However, as Vandall (1990) pointed out, this scenario seems unlikely given that the bulk of the discordant Cretaceous paleomagnetic data has been obtained from intrusions in the Coast Belt, for which direct evidence of paleohorizontal is lacking. Furthermore, geologic evidence indicates that by at least the mid-Cretaceous and possibly earlier the Insular and Intermontane belts were firmly linked (e.g., Monger *et al.* 1982; van der Heyden 1992). In a detailed discussion of the Cretaceous data, Vandall (1990) proposed the Tilt and Moderate Displacement (TMD) model, which invokes post-mid-Cretaceous localized tilts combined with moderate 1000 km northward translation of the combined Coast and Intermontane belts. A similar concept of combined tilt and displacement was proposed by Vandall and Palmer (1990*b*). In addition, Butler *et al.* (1989, 1990) advocate regional tilt as the most significant mechanism responsible for the observed discordance of paleomagnetic directions in the Coast Belt; however, implicit in their interpretation are post-mid-Cretaceous northward displacements approaching 500 km. Alternatively, Umhoefer (1987), Irving and Wynne (1991), and others favor the large-displacement Baja British Columbia model.

These proposed tectonic models provide an important framework within which tests can be carried out to either substantiate, disprove, or further constrain models of Cordilleran tectonics. A recent test of postintrusion tilt for the Spuzzum batholith was carried out by Brown and Burmester (1991) using metamorphic mineral assemblages to estimate paleodepths along a

northeast profile across the batholith. They concluded that post-intrusive tilting of the batholith has taken place. In another study, Irving *et al.* (1992) advanced the discussion of large-scale systematic tilts by demonstrating their occurrence and widespread signature in Tertiary rocks of the Queen Charlotte Islands. Irving *et al.* (1992) used paleomagnetic data from bedded Eocene rocks of the Canadian Cordillera, which are concordant with data from cratonic North America, as a reference datum from which to test for crustal tilt (Bardoux and Irving 1989; Symons and Welling 1989; Irving and Brandon 1990; Vandall and Palmer 1990*a*). Using this datum, discordant magnetizations in Eocene and younger rocks can be interpreted in terms of tilt and (or) rotation. Similarly, westerly tilts in the Eocene extension zone of southeastern British Columbia were identified (Irving and Archibald 1990; Marquis and Irving 1990).

Brown and Burmester (1991) suggested that crustal tilt has occurred within the central-southern Coast Belt domain, which the Spuzzum batholith intrudes (Fig. 1). If post-mid-Cretaceous tilting of rocks within this domain has occurred, younger intrusions may also have been tilted. Time limits for possible tilting in the Coast Belt can be determined by carrying out paleomagnetic studies on younger plutons. To this end the Middle Miocene Rogers Creek pluton was selected for this study (Fig. 1). It is situated northwest of the Spuzzum batholith and intrudes the boundary between the west and central Coast Belt domains (Journeay 1990). Herein new data are reported for the Rogers Creek pluton and combined with a review of the previous Tertiary paleomagnetic data from the Coast Belt.

Geology and sampling

The geologic evolution of the Coast Belt is dominated by Middle Jurassic to Cretaceous plutonism, metamorphism, and deformation. Plutonic rocks of the southern Coast Plutonic Complex areally constitute 60% of the Coast Belt orogen, which has been subdivided into three domains: west, central, and eastern (Journeay 1990). Journeay (1990) has further subdivided the intrusive rocks along the southeast flank of the Coast Plutonic Complex into four magmatic suites on the basis of regional field relationships and geochronology. The Middle Miocene Rogers Creek pluton represents a young granodiorite member of the composite Chilliwack–Pemberton suite of late Tertiary plutonic and related volcanic rocks (Journeay 1990). Within this suite, rocks are locally deformed by high-angle brittle to ductile fault zones, which provide evidence of relatively recent tectonic activity in the southern Coast Belt. The Rogers Creek pluton is situated on the boundary between the west and central domains, which is cut by northeast-trending faults. These faults appear as late kinematic brittle to ductile fault zones within the pluton, the age of which constrains the age limit of motion along these faults. The pluton lies within an uplifted, dominantly Cretaceous, plutonic region that has experienced moderate- to high-grade metamorphism that predates its intrusion. The crystallization of the Rogers Creek granodiorite is 16 ± 1 Ma from U–Pb zircon geochronology (Friedman 1990). Three separate U–Pb rock sample dates were concordant (R. M. Friedman, personal communication, 1992), all of which are in agreement with an earlier K–Ar hornblende age of 16.7 ± 3 Ma (Woodsworth and Roddick 1978).

The pluton was sampled at 16 sites in different phases of the pluton along the Rogers Creek logging road (Fig. 2). Rock types sampled range in composition from fine-grained dioritic xenoliths, which are preferred, to coarse-grained biotite granites

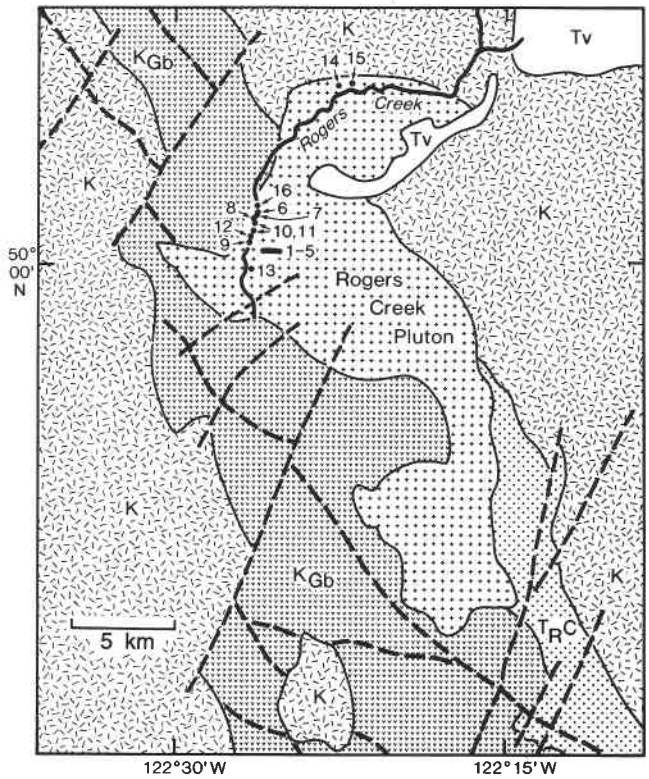


FIG. 2. Rogers Creek pluton paleomagnetic site locations along Rogers Creek. Geology simplified from Journeay (1990). K, Cretaceous plutonic rocks; K_{Gb}, Cretaceous Gambier Group; T_RC, Triassic Cadwallader terrane rocks; Tv, Tertiary volcanics; heavy broken lines are faults.

(Table 1). As will be shown, the granites proved to be poor magnetic recorders.

Methods

At each site, four to seven cores were drilled *in situ* using a portable drill. Cores were oriented *in situ* using both sun and magnetic compasses to detect any possible local magnetic distortions; declinations agreed within a few degrees. In the laboratory, two standard right-cylindrical paleomagnetic specimens were sliced from most cores. Their natural remanent magnetization (NRM) was analyzed using an automated Schonstedt SSM-1 or SSM-2 spinner magnetometer, a Schonstedt TSD-1 thermal demagnetizer, a Schonstedt GSD-5 tumbling alternating field (AF) demagnetizer, and a Sapphire Instruments SI-4 static AF demagnetizer. After initial measurement of NRM, each specimen was subjected to AF or thermal demagnetization techniques in up to 12 steps. AF demagnetization was done up to 100 mT and thermal demagnetization was done up to 700°C.

Stable remanence directions in each specimen were obtained by a joint analysis of stereographic, intensity decay, and orthogonal demagnetization diagrams and least-squares principal component analysis (Kirschvink 1980). Remanent magnetizations were identified as stable end-points (McElhinny and Gough 1963) or as lines on orthogonal plots (Zijderveld 1968; Kirschvink 1980). Directions of remanence were accepted if their maximum angle of deviation (MAD) was $<10°$; the majority were $<5°$. Site mean remanence directions were calculated using two-tiered statistics (Fisher 1953; Watson and

TABLE 1. A- and B-type site mean remanence data

Site	Rock type	UTM coordinates E	UTM coordinates N	M (A/m)	N	D (°)	I (°)	k	R	α_{95} (°)	VGP λ (°N) (Demag. step:	VGP φ (°E) AF/thermal)
1A	fdi	540900	5539300	0.318	5	004	75	104	4.962	7.5	78	246
B						357	74	20	4.803	17.5*		(40/400)
2A	di dike	540600	5539200	0.557	4	050	75	38	3.921	15.7*	61	288
B						043	75	46	3.935	13.7		(40/550)
3A	fdi	540600	5539200	1.231	4	020	71	59	3.949	12.1	77	297
B						024	65	49	3.939	13.3		(20/300)
4A	fdi−gdi	540500	5539200	0.272	7	007	72	166	6.964	4.7	81	262
B						016	75	49	6.878	8.7		(30/400)
5A	fdi−gdi	540200	5539200	0.218	5	354	73	58	4.931	4.9	81	217
B						346	72	30	4.868	14.1		(60/300)
6A	di	540500	5542800	0.237	5	179	−60	115	4.965	7.2	81	062
B						187	−61	90	4.955	8.1		(40/300)
7A	di	540400	5542700	0.205	5	183	−60	56	4.929	10.3	81	043
B						183	−59	43	4.908	11.8		(40/400)
8A	gdi	539900	5541700	1.420	5	178	−59	98	4.959	7.8	79	064
B						178	−59	104	4.961	7.5		(60/400)
9A	bigr	539700	5539600	0.307	5	—	—	—	—	—	—	—
B						021	66	41	4.902	12.1		(20/300)
10A	bigdi	539700	5540500	0.332	5	—	—	—	—	—	—	—
B						037	71	16	4.747	19.8*		(10/200)
11A	gdi	539700	5540800	0.708	5	017	73	159	4.975	6.1	77	281
B						026	75	171	4.977	5.9		(40/500)
12A	bigr	539700	5539800	0.141	5	—	—	—	—	—	—	—
B						033	60	99	4.960	7.7		(10/100)
13A	bigr	539600	5538300	0.147	5	—	—	—	—	—	—	—
B						025	65	33	4.878	13.5		(5/200)
14A	gdi	543800	5547600	0.179	5	166	−68	122	4.967	6.9	81	160
B						162	−68	165	4.976	6.0		(40/500)
15A	gdi	544100	5547700	0.386	5	179	−66	66	4.939	9.5	87	072
B						188	−61	31	4.871	14.0		(40/550)
16A	di	540400	5542700	0.108	5	168	−68	67	4.940	9.4	82	158
B						168	−66	39	4.897	12.4		(40/300)

NOTES: See text for discussion of A- and B-type site means. Rock types as follows: fdi, fine-grained diorite; gdi, granodiorite; di, diorite; bigr, biotite granite; bigdi, biotite granodiorite. M is the mean NRM intensity of magnetization; N is the number of cores; D and I are the site mean declination and inclination of the magnetization given with respect to present horizontal; k is best estimate of Fisher's (1953) precision parameter; R is the length of the resultant vector; α_{95} is the radius of the circle of 95% confidence; and λ and φ are the latitude and longitude of the virtual geomagnetic pole (VGP). Units of AF and thermal demagnetization steps are mT and °C, respectively. * indicates rejected site on the basis of $\alpha_{95} > 15°$.

Irving 1957). These analyses provided A-type site means.

Tectonic analyses of the relative rotations (RR's) (RR < 0 indicates clockwise rotation) and the relative paleolatitudinal displacements (RPD's) (RPD < 0 indicates southward paleolatitudinal displacement) were calculated by the method of Yole and Irving (1980, 1981) and Demarest (1983). The percentage calculation of the present Earth's magnetic field (PEMF) overprints for the Tertiary Coast Belt data (with the exception of the data from Mount Barr) was made by summing the expected direction and its corresponding PEMF direction for each study. During this analysis the expected direction was treated as a unit vector while the magnitude of the PEMF vector was allowed to vary between 0 and 1, until the resultant direction closest to the observed direction was obtained. The proportion of the PEMF direction summed with the expected direction unit vector at that point yields the percentage estimate of PEMF overprint. This analysis requires that the observed mean direction lies close to the plane containing both the expected direction and the PEMF direction, which was the case. This analysis assumes that the divergence of each study's observed direction from its expected direction is due only to PEMF overprints and manifest only on normal polarity sites. As a result, the percentage PEMF overprints may be maximum estimates.

Results

Following detailed AF and thermal step demagnetization, 25% of the specimens exhibited unstable magnetizations and their data were rejected. Specimens from biotite granite and granodiorite (sites 9, 10, 12, and 13, Table 1, Fig. 2) exhibited unstable magnetizations. These magnetizations were rapidly removed (70−95%, in AF's of 5−10 mT), with large directional swings between demagnetization steps (Fig. 3). Similarly, thermal demagnetization of these specimens also demonstrated unstable magnetic behaviour, with distributed unblocking temperatures (T_{UB}'s) combined with large directional changes between steps (Fig. 4). All specimens from these sites have normal magnetizations with steep NRM directions close to the PEMF direction. Therefore, their large soft magnetizations are probably overprints acquired during the present Bruhnes normal polarity chron, and do not record the Earth's magnetic field at the time of cooling.

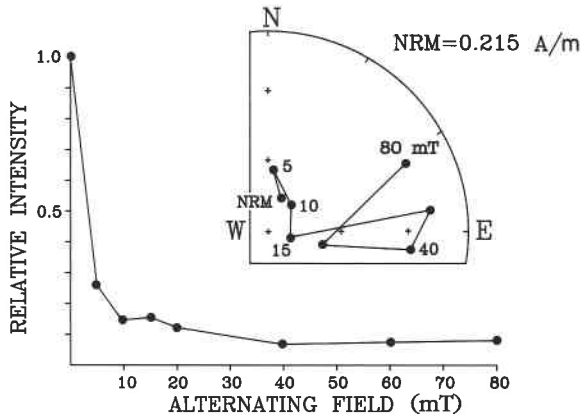


FIG. 3. Example AF demagnetization of an unstable specimen from site 13. Decay of intensity with increasing AF's and the corresponding direction changes illustrated on an equal-area projection. Solid symbols indicate normal polarity directions plotted on the lower hemisphere.

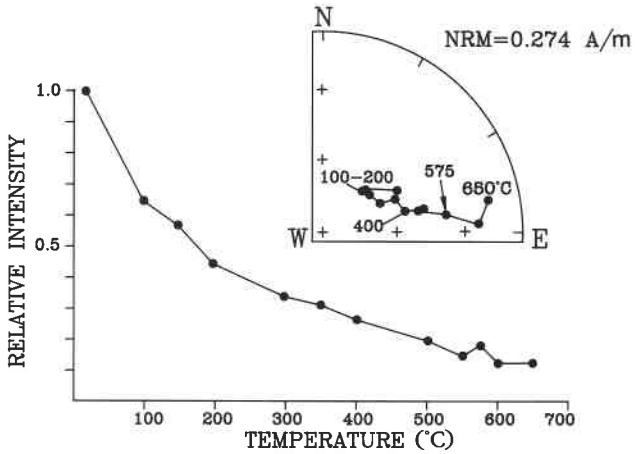


FIG. 4. Example thermal demagnetization of an unstable specimen from site 10. Decay of intensity with increasing temperature and the corresponding direction changes. Symbols as in Fig. 3.

Specimens from the remaining 12 sites are generally more mafic in composition. AF and thermal demagnetization revealed two principal magnetizations, normal and reverse. Most specimens exhibited the initial removal of normal-polarity low coercivity or low T_{UB} magnetizations directed along the PEMF followed by linear univectorial decay (Figs. 5–9). The early-removed magnetizations are thought to be recent Bruhnes overprints.

By 100 mT on AF demagnetization, the magnetic remanence of almost all specimens had been reduced to less than 10% of their original NRM intensity (Fig. 5). The coercivity of the stable component is predominant between 10–40 mT but ranges up to over 100 mT. The most significant variation observed among the AF-demagnetized specimens was the rate at which their intensities decayed, particularly for those of reversed polarity. Reverse specimens are commonly overprinted with a normal magnetization that is removed during step demagnetization. This combination of components leads to an initial drop in intensity followed by a corresponding increase as illustrated in Fig. 7. AF demagnetization examples in Figs. 5, 7, and 8 illustrate the characteristic, stable behaviour of normal,

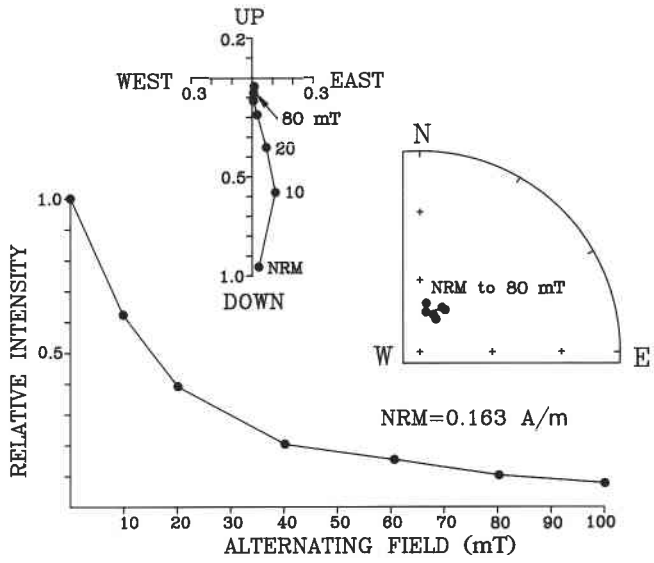


FIG. 5. Example AF demagnetization of a specimen from site 4. Decay of intensity with increasing AF's and the corresponding vertical component vector changes and direction changes. Symbols as in Fig. 3.

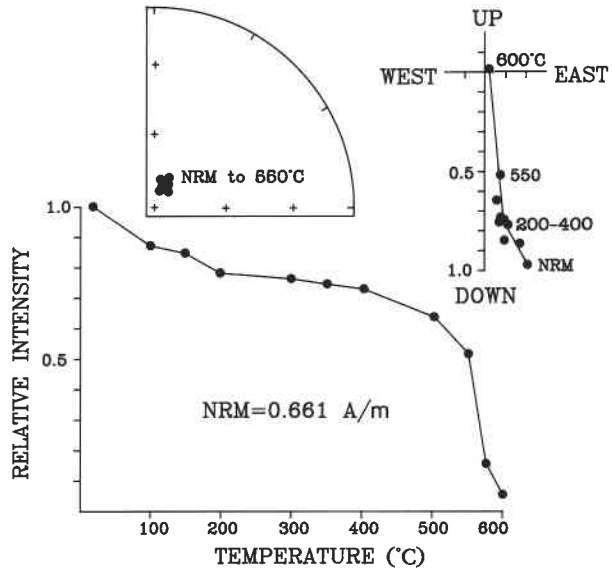


FIG. 6. Example thermal demagnetization of a specimen from site 11. Decay of intensity with increasing temperature and the corresponding vertical component vector changes and direction changes. Symbols as in Fig. 3.

reverse, and mixed polarity specimens from which stable endpoints and least-squares lines could be readily defined.

Thermal step demagnetization also proved effective, with characteristic peak T_{UB}'s in the range 550–570°C (Figs. 6, 9). These observations indicate the magnetite is the principal magnetic carrier of the remanence components. In addition, many specimens exhibited some distributed T_{UB}'s indicating the possible removal of low unblocking temperature viscous remanence magnetizations (VRM's) and (or) multidomain to pseudosingle domain grain magnetizations (Figs. 6, 9). Thermal demagnetization examples in Figs. 6 and 9 illustrate the characteristic, stable, univectorial behaviour of normal and mixed

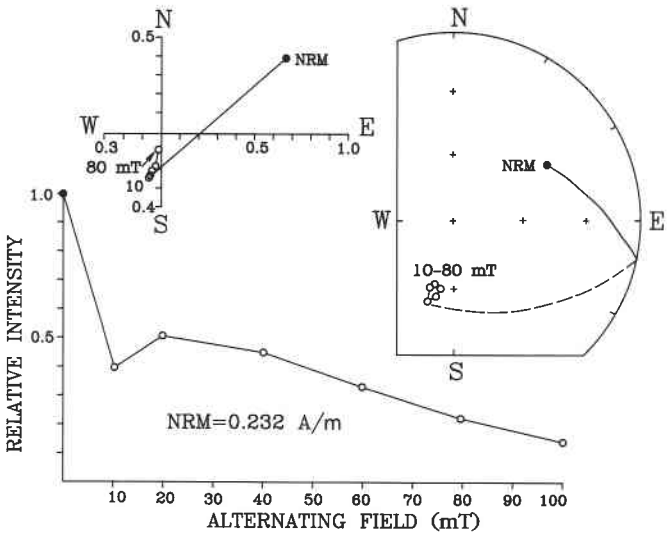


FIG. 7. Example AF demagnetization of a mixed polarity specimen from site 6. Decay of intensity with increasing AF's and the corresponding horizontal component vector changes and direction changes. Solid (open) symbols indicate normal (reverse) polarity directions plotted on the lower (upper) hemisphere.

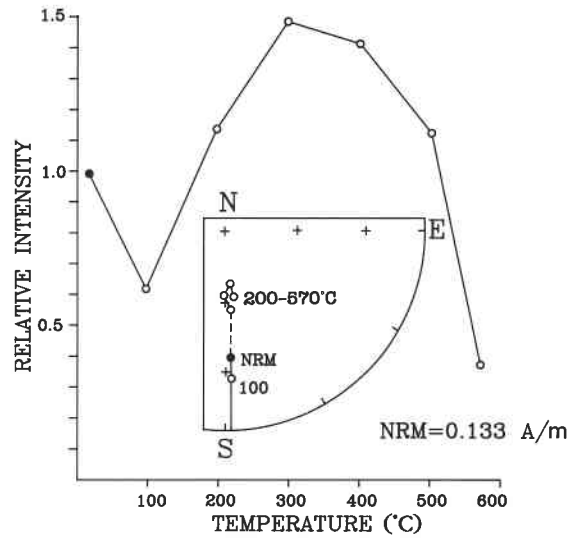


FIG. 9. Example thermal demagnetization of mixed polarity specimen from site 16. Decay of intensity with increasing temperature and the corresponding direction changes. Symbols as in Fig. 7.

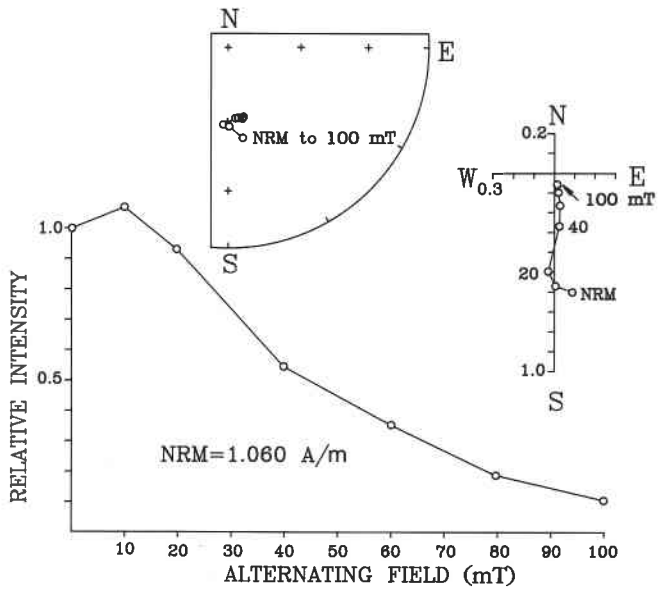


FIG. 8. Example AF demagnetization of a reversed specimen from site 8. Decay of intensity with increasing AF's and the corresponding horizontal component vector changes and direction changes. Open symbols indicate reverse polarity directions plotted on the upper hemisphere.

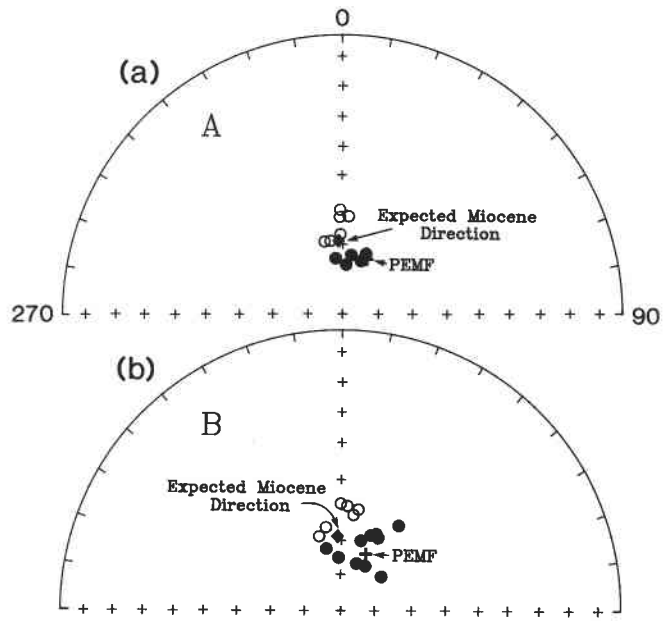


FIG. 10. (*a*) A-type site mean directions. (*b*) B-type site mean directions calculated by minimum site mean dispersion. Solid and open circles are both plotted on the lower hemisphere. Solid circles are north-seeking directions (normal polarity) and open circles are south-seeking directions (reversed polarity inverted through the origin) also on the lower hemisphere, allowing normal and reversed directions to be compared directly. The diamond is the expected Middle Miocene direction. PEMF indicates the present Earth's magnetic field direction.

polarity specimens. In all cases with specimens characterized by superimposed normal and reverse polarity magnetizations, the normal component was systematically removed in relatively low AF's or temperatures, to reveal the stable reverse component (Figs. 7, 9). Stable end-points and least-squares lines were readily defined.

Summary of results

The following discussion is based on A-type site means, which have been determined following the analyses described in the previous methods section. Table 1 is a summary, by site, of the mean paleomagnetic directions and their corresponding virtual geomagnetic poles (VGP's), assigning each core unit weight. No A-type data have been entered in the table for sites 9, 10, 12, and 13, as specimens from these sites failed to retain stable magnetic components. Site 2 has been rejected from further consideration as a result of its large α_{95} ($>15°$). A-type site means are characterized by five normal and six reverse polarity sites and are well grouped (Fig. 10*a*).

TABLE 2. Paleopoles from undisturbed Miocene rock units in North America

Paleopole	Age (Ma)	B	λ (°N)	ϕ (°E)	A_{95} (°)	Reference
Cariboo plateau basalts	10–14	48_f	84.0	219.7	5.1*	Symons 1969*a*
Miocene gabbroic plugs	10–14	17	84.9	213.3	5.2*	Symons 1969*b*
Baja California Peninsula	Miocene	64	86.1	142.6	3.9	Hagstrum *et al.* 1987
Steens basalt	15–18	28_c	88.3	209.0	6.3	Mankinen *et al.* 1987
Columbia River basalt	15–18	59_f	88.7	171.6	4.0	Mankinen *et al.* 1987
Mojave–Sonora volcanics	12–20	65_c	85.5	108.9	4.4	Calderone *et al.* 1990
Reference Pole (1)	20(5–35)	17_p	88	121	3.0	Irving and Irving 1982
Reference Pole (2)	20(5–35)	25_p	85.9	151.1	3.6	Harrison and Lindh 1982
Rogers Creek pluton	16	11	88.3	257.5	5.7	This study
Average of entries 1–6 (K = 565)	10–20	6_p	87.3	182.3	2.8	

NOTES: Age is from the original reference. B is the number of sampling sites ($_f$, flows; $_c$, cooling units; $_p$, paleopoles); λ and ϕ are the latitude and longitude of the paleopole; A_{95} is the confidence limit at P = 0.05 (* indicates an estimate of A_{95} from δp and δm semiaxes error ellipse). The above list of paleopoles includes those studies based on 10 or more sites.

Notable in Fig. 10*a* is the close agreement between the observed site mean directions and the expected Middle Miocene direction, and the approximate antiparallelism of the normal and reverse polarity site mean directions. The normal and reverse polarity component means (normal component D = 008.7°, I = 73.0°, N = 5 sites, α_{95} = 3.3°; reversed component D = 356.3°, I = 63.5°, N = 6 sites, α_{95} = 4.2°) are separated by 10° of arc after inverting the reverse component mean to its north-seeking antipole, permitting direct comparison of the normal and reversed directions. Thus, these data fail the formal reversal test of McFadden and McElhinny (1990), with a critical angle of 5.1°.

Asymmetric reversals are a common feature of the Tertiary paleomagnetic field (e.g., Diehl *et al.* 1988). Possible explanations include the retention of PEMF overprints in measured collections, insufficient averaging of geomagnetic secular variation within the polarity intervals, and different nondipole harmonics associated with each polarity state of the Earth's magnetic field (EMF). As Diehl *et al.* (1988) point out, the systematic far-sidedness of poles calculated from reversed directions relative to poles calculated from normal directions (a feature exhibited by the data from this study) suggests the presence of PEMF overprints or the presence of nondipole components in the time-averaged paleomagnetic field. It is very unlikely that inadequate averaging of secular variation would introduce a systematic effect consistent with both polarities. Furthermore, in this study detailed step demagnetization of specimens yields univectorial decay after removal of "soft" magnetizations (Figs. 7, 8), indicating that significant residual PEMF overprints seem unlikely. Long-term asymmetries in the EMF appear to be the most salient explanation in view of the abundant evidence suggesting the existence of biased nondipole components for perhaps the last 100 Ma (e.g., Merrill *et al.* 1979; Coupland and Van der Voo 1980). The asymmetric reversal described here is consistent with the asymmetric Miocene paleomagnetic field recorded elsewhere in North America and does not appear to manifest significant PEMF overprints.

The above analyses indicate that in all probability secular variation has been averaged within error and that the site means are primary magnetization components for the Rogers Creek pluton, which were acquired during cooling following crystallization at 16 Ma. Given the concordant 16 Ma U–Pb and K–Ar ages for the pluton and the presence of reversals, 16 Ma in all probability also dates the age of magnetization.

The mean Rogers Creek pluton paleomagnetic direction from the 11 accepted A-type site means (Table 1) yield a unit mean directed at D = 000.7°, I = 67.9° (N = 11, k = 138, α_{95} = 3.9°). The Rogers Creek pluton paleopole calculated from the site VGP's (Table 1) is listed in Table 2.

Comparison with North American data

Miocene paleopoles between 10 and 20 Ma in age from North America outside regions of post-Miocene tectonic disturbance are listed in Table 2. Paleopoles derived from fewer than 10 sites have been excluded. Figure 11 illustrates their good agreement clustering near the Earth's geocentric axial dipole (GAD). The average of these paleopoles within the 10 Ma window yields a 10–20 Ma North American reference pole with very low error (A_{95} = 2.8°; last entry, Table 2). Previously determined North American reference poles, calculated by a 30 Ma moving average window centred about 20 Ma (Harrison and Lindh 1982; Irving and Irving 1982) are in close agreement with this pole even though they are calculated with a larger averaging window and do not include more recent data (Table 2). This emphasizes the high accuracy of the late Tertiary paleopole positions and demonstrates the relatively slow rate of apparent polar wander during the middle to late Tertiary.

Figure 12 illustrates the excellent agreement between the Rogers Creek pluton pole and the 10–20 Ma North American reference pole from this study. This agreement indicates that the Rogers Creek pluton has not been displaced relative to stable North America or tilted significantly since intrusion.

Tertiary Coast Belt data

The following discussion of Tertiary Coast Belt data is constrained by concordant paleomagnetic results from Canadian Cordilleran bedded Middle Eocene rocks (Table 3). These results provide a reference datum from which to examine crustal tilt because they imply that the major elements of the Canadian Cordillera were in their present position relative to North America by Mid-Eocene time. This follows from the good agreement among the Canadian Cordilleran Middle Eocene poles and the available Middle Eocene cratonic poles (Fig. 13). Tectonic analysis of the Middle Eocene poles from the Canadian Cordillera (Table 3) with respect to the average of the three Middle Eocene North American cratonic poles (entry 4, Table 3) shows a negligible relative displacement overall (RR = 2°

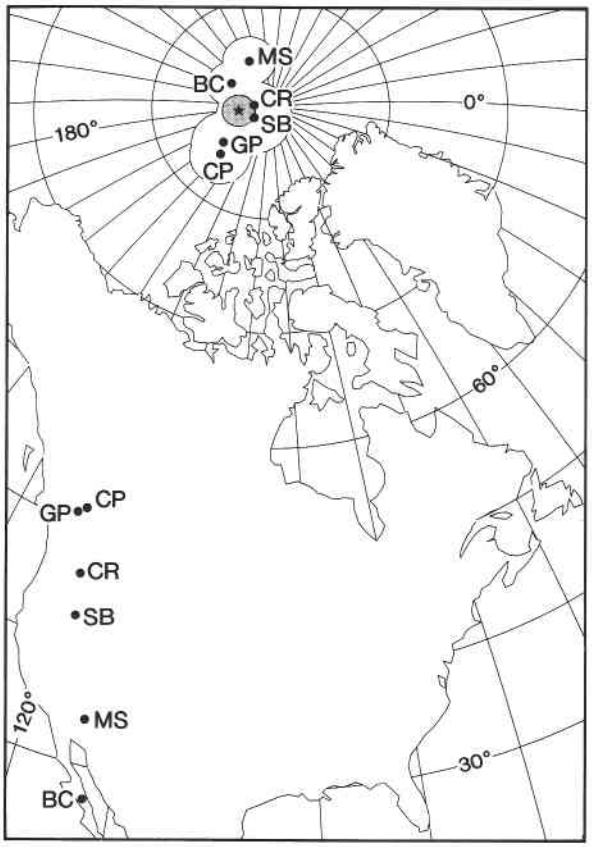


FIG. 11. Miocene localities and paleopoles from tectonically undisturbed regions of North America, and the corresponding mean reference pole (star). Poles from Table 2. CP, Cariboo plateau basalts; GP, gabbroic plugs; BC, Baja California peninsula; SB, Steens basalt; CR, Columbia River basalt; and MS, Mojave–Sonora volcanics. Standard errors (P = 0.37) are drawn for this figure and pole Figs. 12–14.

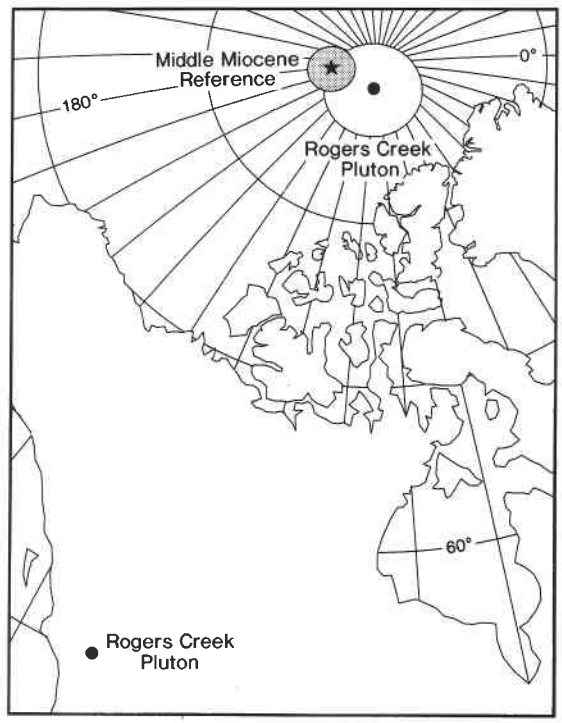


FIG. 12. The paleopole from Rogers Creek pluton compared with the 10–20 Ma reference paleopole determined in this study for North America (star).

clockwise, RPD = 0°). This provides the basis for calculating a new Middle Eocene North American reference pole by including data from the Canadian Cordillera (48–52 Ma, last entry, Table 3). Thus it is unlikely that discordant Canadian Cordilleran magnetizations in post-Eocene rocks are the result of latitudinal displacement relative to North America.

Given this constraint, the tectonic analyses of Tertiary Coast Belt paleomagnetic data were calculated with respect to their applicable reference poles established in this study. Table 4 lists the seven Tertiary paleomagnetic studies from the Coast Belt with their corresponding RR's and RPD's. The Middle Miocene and Middle Eocene reference poles used are outlined in Tables 2 and 3, respectively. The Late Eocene to Early Oligocene pole was calculated from 46 cooling units from the Mogollon–Datil volcanic field, southwestern New Mexico (30–38 Ma K–Ar ages from Table 1, Diehl *et al.* 1988) and 27 flow means from the Indian Well Formation, northeast Nevada (30–37 Ma K–Ar ages from Table 1, tuffs of Hackwood and Jiggs, Palmer *et al.* 1991). This reference pole (30–38 Ma) is positioned at 84°N, 147°E (N = 73, A_{95} = 4.3°).

In all cases, except for the Chilliwack batholith for which no pole error was given by Beck *et al.* (1982), the RR's and RPD's are within or close to 95% error limits. This demonstrates that all individual rotations and displacements are not significantly different from zero, indicating that these poles are concordant with their respective reference poles with the exception of the Mount Barr pluton pole (Table 4). By contrast, the combined Tertiary results exhibit systematic clockwise rotations and overall southward displacements relative to North America (Table 4), the average being RR = 8° clockwise and RPD = 3° southward, excluding the Mount Barr pole (including Mount Barr increases the average to RR = 10° clockwise and RPD = 4° southward). Figure 14 illustrates the systematically clockwise near-sided Tertiary poles. Given that a southward displacement of ~300 km since magnetization can be reasonably precluded, these poles could be ascribed to regional-scale downward tilting to the northwest. Although evidence of a systematic, regional style of tilt has been documented in the Queen Charlotte Islands by Irving *et al.* (1992), it seems very unlikely for the Coast Belt, as Symons's (1974, 1977*b*) studies of the Hawksbury warp would indicate. A more cogent explanation is the incomplete removal of PEMF overprints, which would bias the magnetic directions towards the present Earth's magnetic field direction.

All previous Tertiary Coast Belt magnetizations have been obtained from the analysis of AF step-demagnetized pilot specimens (usually 1 or 2 detailed pilot specimens per site) and optimum bulk AF demagnetization (usually at 1 or 2 steps) combined with minimum site mean dispersion techniques. Thermal demagnetization was not used except for a few specimens from the north-northwest Hawksbury warp plutons (Symons 1974, 1977*b*) and the Chilliwack batholith (Beck *et al.* 1982). These analyses require that the coercivity spectra for all specimens within a site must be the same so that stable end-points are reached for all specimens within the site at the same demagnetization step. If there is within-site variation of coercivity spectra and PEMF overprints are present, they will not be sufficiently removed by these demagnetization–analysis tech-

TABLE 3. Middle Eocene paleomagnetic poles from bedded rock units in North America including British Columbia

Paleopole	Age (Ma)	B	λ (°N)	ϕ (°E)	A_{95} (°)	RR (°)	RPD (°)	Reference
Absaroka volcanics	48	19	83.5	177.4	10.1	—	—	Shive and Pruss 1977
Bearpaw Mountains volcanics	48	30	82.0	188.9	6.2	—	—	Diehl *et al.* 1983
Highwood Mountains volcanics	50	30	82.1	163.0	7.0	—	—	Diehl *et al.* 1983
Average of entries 1−3 (K = 1780)	48−50	3_p	82.7	176.4	2.9	—	—	
Kamloops Group volcanics	49	24	81.4	221.6	11.6*	+6±19	−5±10	Symons and Wellings 1989
Kelowna and Castlegar volcanics	52	28	85	197	10.0	+3±14	0±8	Bardoux and Irving 1989
Ootsa Lake volcanics	50	13	88.0	354.6	11.5	−13±16	+4±9	Vandall and Palmer 1990*a*
Flores volcanics	50	12	81.1	188.0	9.9	−5±14	−1±8	Irving and Brandon 1990
Average RR and RPD for British Columbia						−2	0	
Average of 1−3 and 5−8 (K = 331)	48−52	7_p	84.1	190.6	3.3	—	—	

NOTES: Age is from the original reference. B is the number of sampling sites ($_p$, paleopoles); λ and ϕ are the latitude and longitude of the paleopole; A_{95} is the confidence limit at P = 0.05 (* indicates an estimate of A_{95} from δp and δm semiaxes error ellipse); RR and RPD are the relative rotations and relative paleolatitudinal displacements, respectively. The above list of paleopoles includes those studies based on 10 or more sites.

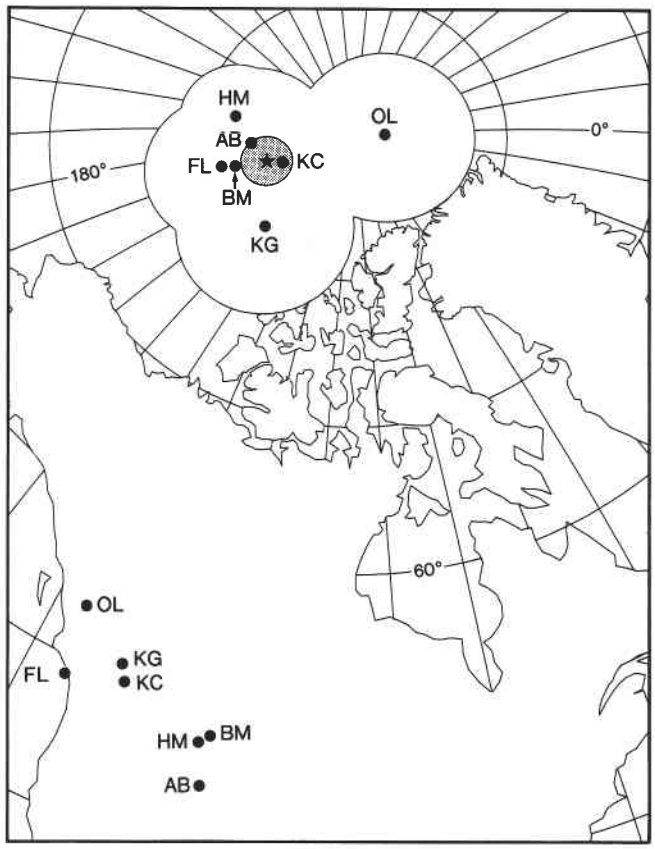


FIG. 13. Middle Eocene localities and paleopoles from bedded volcanic rocks of North America and their corresponding mean reference pole from this study (star). Poles from Table 3. AB, Absaroka volcanics; BM, Bearpaw Mountains volcanics; HM, Highwood Mountains volcanics; KG, Kamloops Group volcanics; KC, Kelowna and Castlegar volcanics; OL, Ootsa Lake volcanics; and FL, Flores volcanics.

niques. To demonstrate the potential bias from insufficiently removed PEMF overprints, these techniques were simulated for a second analysis of the Rogers Creek pluton data. In this analysis specimen directions were selected using minimum site mean dispersion at an optimum demagnetization step chosen for each site in the range 5−60 mT for AF and from 100 to 550°C for thermal (B-type site means; Table 1). Although each specimen in the current study was step demagnetized in detail, this analysis simulates the site means that would have been determined employing the less rigorous techniques generally used in previous studies. For demonstration, data rejected outright in the A-type analysis for unstable sites 9, 10, 12, and 13 are included for B-type site means because, fortuitously, reasonable site mean precisions (average k = 47) were calculated. This illustrates how data from unstable specimens can appear to provide acceptable site means without detailed AF and thermal step demagnetization data required to demonstrate their poor magnetic stability.

The B-type site mean α_{95}'s (Table 1) exhibit an overall average increase of 3.2° from the A-type site means, demonstrating the greater error in this second analysis. Fortuitously, sites 2, 8, 11, and 14 yield slightly lower α_{95}'s; however, site 2 does not pass the A-type selection criteria ($\alpha_{95} < 15°$) and sites 8, 11, and 14 show insignificant decreases in α_{95}'s. These are examples of sites that have similar coercivity spectra for all specimens within the site so that stable end-points are reached at the same demagnetization step. Therefore, minimum site mean dispersion techniques are valid for sites 8, 11, and 14. However, this is not the case for the remaining B-type sites, which show an overall large average increase in α_{95}'s of 8.3°. The plot of the B-type site means (Fig. 10*b*) illustrates the bias from the downward-directed PEMF, which steepens (shallows) the inclinations of the normal (reverse) polarity sites toward the southeast (northwest). The B-type unit mean calculation (D = 13.5°, I = 71.1°, k = 25, α_{95} = 8.2°, N = 14) excluding those sites with $\alpha_{95} > 15°$, shows a large increase in dispersion (A-type k = 138) and a strong bias towards the PEMF (D = 021°, I = 72°). The greater number of normal (8) versus reverse (6) polarity sites is significant because the PEMF bias would tend to cancel out if the ratio of normal to reverse sites was even. However, this assumes that the optimum demagnetization steps for normal versus reverse sites are similar such that the magnitude of PEMF overprints are the same in normal and reverse specimens. In practice this is rarely true because the normal PEMF directions are readily distinguished from the reverse magnetizations, and commonly larger AF's or thermal demagnetization temperatures are used to isolate reverse magnetizations, thereby more effectively removing PEMF overprints from reversed specimens.

It is reasonable to conclude from these observations that

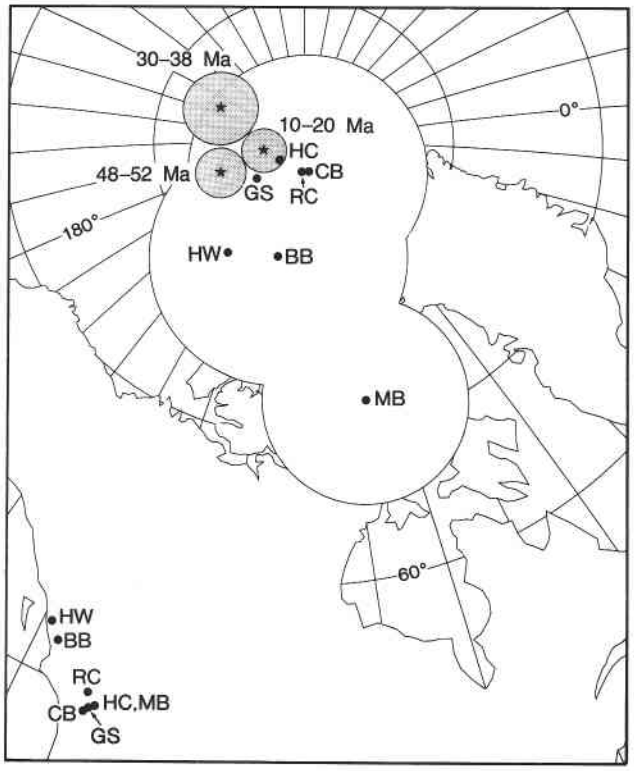


FIG. 14. Tertiary paleopoles from Coast Belt intrusions and their corresponding mean reference poles from this study (stars). Poles from Table 4. BB, Brown basalt dikes; RC, Rogers Creek pluton; MB, Mount Barr pluton; GS, Grotto and Snoqualmie batholiths; CB, Chilliwack batholith; HC, Hope Complex; and HW, NNW Hawksbury warp plutons.

PEMF overprints have been insufficiently removed from the Rogers Creek data when the site mean directions are determined by the minimum site mean dispersion method (B-type site means). This comparison demonstrates that A-type site means (Fig. 10*a*) are better grouped (A-type $k = 138$ vs. B-type $k = 25$) and exhibit negligible PEMF overprints relative to B-type site means (Fig. 10*b*), which show a strong PEMF overprint bias as well as greater dispersion. Furthermore, this comparison highlights the improved resolution of the principal magnetization vectors resulting from selecting directions for specimens individually by thorough step demagnetization and independent of the coercivity or unblocking temperature spectra of companion specimens within the same site. Given that the PEMF and GAD directions are relatively close to the Cretaceous to Recent paleomagnetic directions for North America, it is inherently difficult to separate discretely these magnetizations without the use of detailed AF and thermal step demagnetization on all specimens.

It is evident from this exercise that the observed clockwise near-sided Tertiary Coast Belt poles can be ascribed to partial hybrid magnetizations that include magnetic components aligned along the PEMF direction. Consistent with this interpretation is the observation that those poles determined from the largest percentages of normal polarity sites exceeding 75% (Mount Barr pluton, Brown basalt dikes, Hawksbury warp plutons) also exhibit the greatest near-sided bias towards the PEMF (Table 4, Fig. 14). By assuming the divergence of observed directions from expected directions is due only to PEMF overprints, maximum estimates of the right-handed and near-sided

TABLE 4. Summary of Tertiary Coast Belt data from intrusives

Paleopole	Age (Ma)	*B*	*N* (%)	λ (°N)	φ (°E)	A_{95} (°)	RR (°)	RPD (°)	Reference
Brown basalt dikes	0–10	9	78	82	248	16.2*	−8±28	$-6\pm13_a$	Symons 1968
Rogers Creek pluton	16	11	45	88	258	5.7	−4±8	$0\pm5_a$	This study
Mount Barr pluton	16–21	5	80	72	274	14.0*	−25±25	$-11\pm11_a$	Symons 1973
Grotto and Snoqualmie batholiths	26/15	7	57	86	209	5.0*	−9±7	$-1\pm4_a$	Beske *et al.* 1973
Chilliwack batholith	30–36	34_s	0	88	268	N.A.	−12±?	$1\pm?_b$	Beck *et al.* 1982
Hope Complex	35–41	5	60	88	208	5.6*	−8±9	$-5\pm5_b$	Symons 1973
NNW Hawksbury warp plutons	40–50	11	82	81	226	8.0	−5±15	$-6\pm7_c$	Symons 1974, 1977*b*
Overall percentage of normal polarity sites			63						
Average RR and RPD excluding Mount Barr							−8	−3	
Average RR and RPD including Mount Barr							−10	−4	

NOTES: Age is from the original reference. *B* is the number of sampling sites ($_s$, samples); % *N* is the percentage of normal polarity sites; λ and φ are the latitude and longitude of the paleopole; A_{95} is the confidence limit at $P = 0.05$ (* indicates an estimate of A_{95} from δp and δm semiaxes error ellipse); RR and RPD are the relative rotations and relative paleolatitudinal displacements, respectively. The west-northwest-tilted data from the Hawksbury warp has been excluded. Subscripts $_{a, b}$, and $_c$ correspond respectively to the following reference poles used: 10–20 Ma (87.3°N, 182.3°E, $A_{95} = 2.8°$, Table 2); 30–38 Ma (84°N, 147°E, $A_{95} = 4.3°$, see text); and 50 Ma (84.1°N, 190.6°E, $A_{95} = 3.3°$, Table 3).

bias of the Tertiary poles from the Coast Belt (RR = 8° clockwise and RPD = 3° southward), excluding the Mount Barr pluton pole, reflect the alignment of 10–25% of their remanence components along the PEMF (normal polarity sites). The percentage of PEMF overprints for the Mount Barr pluton pole is possibly quite large; however, this is uncertain because its position is even near-sided and clockwise of the PEMF pole. VRM may have been acquired throughout the Brunhes, therefore not all of the remanence components are aligned in the PEMF direction. Owing to the low number of sampling sites (5) for the Mount Barr pluton and the very large A_{95} of 14° for its pole, this low accuracy result is likely complicated by more than just PEMF overprints.

Each Tertiary Coast Belt study defines its pole with reasonable accuracy (with the exception of the Mount Barr pluton), sufficient for the needs of the original studies. The PEMF bias becomes more apparent when these results are compared directly, and as a result they complicate the investigation of Coast Belt tectonic displacements. It is clear from the analysis of these paleomagnetic data that no tectonic tilt or displacement of the Coast Belt can be inferred. These observations serve as a case study of the effect of PEMF overprints that may affect other paleomagnetic studies to various degrees and are, as of yet, not identified. Within an individual study, the effect of overprints can be within error; however, it has been shown that a collective systematic bias, incorrectly interpreted, could be used as evidence for displacements or tilts when, in fact, the origin of the bias is rock magnetic rather than tectonic. Given the tectonic problems that paleomagnetism is currently being used to study, with inherently smaller displacements, workers are attempting to push the limits of the paleomagnetic technique. As a result, this analysis demonstrates the need for thorough detailed AF and thermal step demagnetization of all specimens, to maximize the paleomagnetic directional information needed to ensure that overprints are sufficiently removed.

Tectonic implications and summary

The paleopole determined for the Rogers Creek pluton is concordant with the North American reference pole. This provides a negative test for post-Miocene tilt of the Rogers Creek pluton, indicating that it has not been displaced since intrusion at 16 Ma. Combining this result with similarly concordant Tertiary results from the Hope Complex (Symons 1973), Chilliwack batholith (Beck *et al.* 1982), and the Grotto and Snoqualmie batholiths (Beske *et al.* 1973) to the south (Fig. 1), all of which exhibit minimal PEMF bias, delineates a southern Coast Belt domain within which no evidence of post-Middle Miocene tectonic tilting has been observed. This observation does not support or preclude the possible tilting of the mid-Cretaceous Spuzzum batholith within this domain. Therefore, no further implications for Cretaceous Coast Belt paleolatitudes that would bear on the tectonic models of Monger *et al.* (1982) or van der Heyden (1992) have been established. This concordant result does, however, provide a minimum age limit for the proposed Spuzzum batholith tilt (Brown and Burmester 1991) and a Middle Miocene paleohorizontal datum for future work investigating crustal tilts. Further paleomagnetic investigations of crustal tilt will play a critical role in understanding Cordilleran tectonics, as it is the best method to detect and quantify tectonic block motions.

Van der Heyden's (1992) Andean subduction model for the Coast Belt describes Middle Jurassic accretion of a single composite superterrane and subsequent formation of the Coast Belt as a result of prolonged subduction. Some form of the tilt hypothesis appears to be the most consistent scenario with this model. Indeed, van der Heyden (1992) favours the tilt and moderate displacement hypothesis outlined by Vandall and Palmer (1990*b*) and described in detail by Vandall (1990). Although the negative tilt test for the Rogers Creek pluton fails to provide direct evidence bearing on the discussion of Coast Belt tilt and the Andean subduction model, this analysis of Tertiary paleomagnetic data provides an important framework within which paleomagnetic data from Cretaceous Coast Belt intrusions can be discussed. Studies of Cretaceous to early Tertiary rock units are still needed to resolve the paleolatitudes for the Coast Belt during this period.

The observation of a systematic PEMF bias in the analysis of Tertiary paleomagnetic data serves as an example of Bruhnes overprints that may affect other paleomagnetic results. As a consequence, a case is made for the importance of thorough step demagnetization of all specimens in future studies, combined with the use of both AF and thermal demagnetization techniques.

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